9. Conceptual Framework and Research Agenda for Mechanism-Aware, Risk-Adaptive Patient-Level Protection

Sections 4–8 synthesize auditing, disparity, candidate drivers, mitigations and trade-offs. **This section is an author-derived conceptual framework and research agenda, not an existing validated clinical pipeline.** It links reliable patient-level vulnerability estimation to persistent exposure, causal mechanism tests, targeted protection and independent re-auditing (Figure 9.1). Each transition states a hypothesis requiring clinical validation.

```
flowchart TD
    A["Aggregate audit"] --> B["Patient-level audit"]
    B --> C["Persistent exposure"]
    C --> D["Mechanism test"]
    D --> E["Targeted protection"]
    E --> F["Independent re-audit"]
    F -. "material update" .-> B
```

Figure 9.1. Proposed research sequence from aggregate screening to patient-level evidence and selective protection. Persistent exposure, the shared/specific mechanism, and intervention selection are hypotheses; independent re-auditing tests benefit, transfer and utility after protection. An editable vector version accompanies this section.

9.1 Reliable Patient-Level Vulnerability Estimation

An audit-specific value such as the example below requires a declared membership event, patient unit, attack access, operating point and repeated IN/OUT observations [6]. Knolle et al.'s direct clinical result illustrates heterogeneity under one repeated-training design; it is not a context-free probability of privacy harm [3]. The framework treats c as an explicitly recorded configuration rather than hiding these choices behind a single score.

$$V_{p,c} = 0.87$$

This value is illustrative only; its metric and scale, configuration c, and sampling uncertainty must be declared before it can be interpreted.

Sample-specific reference distributions and difficulty calibration distinguish training-dependent behavior from intrinsic predictability [2,13]. Song and Mittal's posterior membership score additionally depends on a stated prior and estimated conditional distributions [15]. A patient with several records requires a joint or prespecified aggregate statistic calibrated at the patient unit; a maximum of record scores alone does not control patient FPR. Conceptually,

$$\text{Reliable vulnerability} \neq \text{raw confidence or loss.}$$

Instead, vulnerability should reflect deviation from the behavior expected without the target patient's contribution. For multi-record patients, that counterfactual must be defined at patient level; calibrating records separately and then taking their maximum does not automatically produce a calibrated patient-level false-positive rate.

No single attack provides an attack-independent ground truth. Threshold attacks, LiRA, RMIA, and confidence- or margin-based procedures exploit related but non-identical signals and can differ in assumptions, computational budgets, operating points, and sensitivity to reference mismatch [2,6]. RMIA introduces a likelihood-ratio attack that combines reference models with population samples and achieves strong attack power with comparatively few reference models across several evaluated settings [48]. This makes it useful for practical stress testing, but its evidence is primarily record-level and non-clinical and does not establish that RMIA alone identifies persistently vulnerable patients. Disagreement between attacks may reflect estimation failure, or it may arise because different attacks instantiate different feasible adversaries. A future audit could therefore preserve an attack-indexed patient profile,

$$\mathbf{V}_p = \left(V_p^{\text{loss}}, V_p^{\text{LiRA}}, V_p^{\text{RMIA}}, \ldots\right),$$

and evaluate rank correlation, threshold agreement, and high-risk-set overlap across attacks that share a compatible membership hypothesis. Combining heterogeneous scores into one number would require explicit calibration to a common estimand rather than simple averaging.

Repeated patient inclusion and exclusion, target/reference training and patient-cluster uncertainty are needed for precise tail estimates. A global threshold need not control individual FPR, and finite reference populations can bias efficient estimates; Jälkö et al.'s findings remain a methodological preprint [14]. The minimum protocol is consolidated in Table 4.1.

The research gap is persistence of the **same** multi-record patient across compatible configurations, rather than the existence of heterogeneous scores [3,33]. A prespecified profile could retain:

$$\mathcal{P}_p = \left\{\left(\widehat{V}_{p,c}, CI_{p,c}\right) : c \in \mathcal{C}\right\}.$$

Here c indexes compatible attacks and training conditions. A persistence rule could count configurations in which a lower confidence bound exceeds a **metric-specific** threshold, but should never average incomparable AUC, likelihood-ratio and posterior scales. This proposed rule needs external calibration and validation.

9.2 Why Are Some Patients Persistently High-Risk?

Persistence means recurrent vulnerability across prespecified compatible models and attacks with uncertainty controlled; it is not an intrinsic patient trait. Section 5 documents extreme patient-linked exposure, while general benchmarks show sample heterogeneity [15] and relative memorization under changed data composition [35]. None proves that patient identities remain in the tail across all configurations.

Long-tail theory links atypical examples to selective memorization [27,28], and fixed-target analysis permits heterogeneous leakage [33]. Section 5 documents tail enrichment in some underrepresented clinical groups, but group size, disease rarity and local density differ. A matched clinical test should ask whether persistently exposed patients have fewer similar neighbors after controlling diagnosis, institution, record count and performance.

Whole-patient deletion $D \setminus \mathcal{R}_p$ can test what inclusion changes in predictions and attacker-visible statistics. Influential examples can leak even when a model generalizes well [32], but conventional local approximations can fail for groups of correlated records [39,40]. Therefore,

$$\text{Influence} \neq \text{privacy vulnerability}.$$

The unresolved question is whether recurrent exposure tracks a stable, privacy-relevant component of influence rather than influence magnitude alone.

Persistence may also depend on the structure of a patient's contribution. Repeated records change training weight and can carry a shared signature; near-duplicate images differ from diverse visits, and the attacker may hold a different number of records than the model saw. Dependencies can alter attacks [23], but a monotonic effect of record count is unproven. The clinical model comparison in Section 6 suggests an expanded tail for larger configurations, whereas regularization can change the capacity relationship [36,37]. The relevant hypothesis concerns interactions among data, training and attack, rather than a fixed patient trait.

The hypothesis is whether recurrent exposure corresponds to a stable patient-specific contribution. Matched comparisons and whole-patient deletion can test it; Section 9.3 develops the proposed mechanism.

9.3 From Association to Mechanism

Rarity, record count, loss and influence co-vary; association alone cannot identify which patient-induced change creates an attack signal. A causal clinical test must show that including $\mathcal{R}_p$ changes an attacker-observable statistic, that the difference separates IN from OUT models, and that a targeted intervention reduces this difference while controlling alternative explanations.

Possible pathways include

$$\text{rarity} \rightarrow \text{patient-specific learning} \rightarrow \text{membership distinguishability},$$

or

$$\text{multiple records} \rightarrow \text{greater training influence} \rightarrow \text{membership distinguishability}.$$

These pathways are hypotheses; the patient-induced model change must be measured and selectively perturbed before a causal interpretation is warranted.

The most direct starting point is a patient-level inclusion-exclusion experiment. Let $\mathcal{R}_p$ denote all records contributed by patient p, let D be the complete training dataset, and define

$$D_{-p} = D \setminus \mathcal{R}_p.$$

Compare f_D with $f_{D_{-p}}$ over repeated paired training runs, removing **all** records of patient p. Measure losses, logits, representations, clinically similar test cases and attack statistics separately. A single model pair cannot distinguish a patient effect from optimization randomness [28,32].

Clinical influence is a change in useful prediction; privacy-relevant influence is a change that separates membership states. A rare patient's contribution may improve generalization without a strong MIA signal, or create localized leakage with little effect on aggregate accuracy. Influence-function approximations concern local upweighting and can misestimate grouped deletion [39,40]. The shared/specific hypothesis below seeks a component that carries membership evidence without discarding clinically useful learning.

A testable hypothesis is that the patient's learned contribution contains both shared and patient-specific components. Conceptually,

$$g_p = g_p^{\text{shared}} + g_p^{\text{specific}},$$

where g_p^{shared} represents information also supported by clinically similar patients and g_p^{specific} represents a residual change attributable more distinctively to patient p. For example, imaging features characteristic of a rare disease may constitute shared clinical information even when only a small subgroup carries them, whereas idiosyncratic anatomy, acquisition artifacts, repeated near-duplicates, or an unusual combination of attributes may contribute a more patient-specific residual. Long-tail learning provides a motivation for this distinction because rare examples can require selective memorization and estimated memorization and influence are highly heterogeneous [27,28]. However, the proposed decomposition is presently a conceptual hypothesis, not an established or uniquely identifiable mathematical separation. A residual defined by one representation, neighborhood, or gradient metric may differ from that defined by another. The hypothesis to test is therefore not that every patient-specific residual constitutes leakage, but whether a stable residual predicts persistent vulnerability across compatible models and attacks:

$$g_p^{\text{specific}} \xrightarrow{?} \text{ persistent membership distinguishability}.$$

A staged experiment would identify recurrently vulnerable patients, estimate rarity, multiplicity, influence and candidate residuals under repeated whole-patient inclusion/exclusion, then test a targeted attenuation in held-out audit runs. Multivariable prediction alone remains correlational; the stronger claim requires a repeated, selective intervention that lowers membership distinguishability while

preserving clinical generalization. The shared/specific split is only a hypothesis and may depend on representation and neighborhood definitions.

9.4 Targeted Mitigation of Extreme-Risk Patients

Uniform DP-SGD supplies an important baseline safeguard under a declared adjacency relation [8,41]. Yet a small clinical tail can remain much more empirically distinguishable than the population mean [3], and nonclinical studies show heterogeneous sample-specific scores and possible subgroup utility costs [15,47]. This motivates testing **additional** selective protection against a comparable uniform baseline; superiority has not been established. The intervention question is whether resources can be concentrated on patients with reproducible exposure while maintaining the baseline guarantee and clinically important information across the full population. A targeted arm and a uniform arm should therefore be specified before looking at protected-model outcomes, with comparable protected units, accountants, contribution limits and either utility or resource constraints.

Targeted patient-level protection requires a predefined mapping from reliable audit evidence to an intervention. The baseline high-risk set and threshold should be fixed as specified in Section 8.2 before the protected model is evaluated. Eligibility for stronger intervention should reflect uncertainty and the persistence criteria developed in Sections 9.1 and 9.2 rather than a point estimate alone. A general policy may be written as

$$a_p = g\Big(\{\widehat{V}_{p,c}, CI_{p,c}\}_{c\in\mathcal{C}}, S_p\Big),$$

where a_p is the assigned intervention, $CI_{p,c}$ quantifies uncertainty, and S_p summarizes prespecified stability across compatible audits. Eligibility could require a lower confidence bound above a metric-specific threshold in more than a prespecified fraction of compatible runs; ambiguous patients could receive the baseline safeguard while further audit data are gathered. Persistently vulnerable patients might receive stricter contribution bounds, lower sampling frequency, stronger clipping, or modified weights. Individualized DP-SGD shows that examples can receive different sampling or clipping parameters, but its budgets arise from preferences or policy and its analysis is example-level [45]. REPS uses estimated outlier characteristics to alter record influence during private synthetic-data generation, but remains a record-level tabular proof of concept [46]. Training-data-derived screening, selection and any released scores or assignments must be included in an end-to-end formal privacy analysis if DP is claimed; a post hoc empirical benefit cannot repair missing accounting.

Persistently vulnerable patients may also carry rare and clinically valuable information, so targeted protection should not be reduced to "more risk, more noise." Indiscriminately suppressing a contribution could discard the signal needed to learn uncommon conditions. Section 9.3 instead motivates the schematic hypothesis

$$g_p = g_p^{\text{shared}} + g_p^{\text{specific}}, \qquad \tilde{g}_p = g_p^{\text{shared}} + \alpha_p g_p^{\text{specific}}, \qquad 0 \le \alpha_p \le 1.$$

Under this hypothesis, $\alpha_p < 1$ would attenuate a candidate patient-specific component while preserving information shared with clinically comparable patients. Long-tail memorization and influence studies make the distinction scientifically plausible [27,28], but do not provide an operational separation between clinically useful and privacy-relevant gradients. One experiment could estimate a candidate residual on discovery runs, freeze the decomposition and policy, and compare its attenuation with whole-patient clipping, uniform noise and a sham intervention in held-out runs. It should measure both membership distinguishability and predictions for similar rare-condition patients: a fall in vulnerability accompanied by loss of a shared clinical signal would fail the proposed mechanism's purpose. The decomposition, its estimators, and the mapping from audit evidence to α_p require experimental validation. This is a research hypothesis—not an identified quantity, established algorithm, or formal privacy guarantee.

The intervention could reveal a new vulnerable layer [35], impair utility for rare conditions [47], or misallocate protection because the screening estimate is uncertain [7]. Selection errors have different consequences: false negatives miss additional protection, while false positives may weaken

the contribution of a patient whose information is useful. The assigned label, budget or mechanism should be treated as sensitive metadata. An independent audit should include selected patients, unselected patients and random controls, with a fixed membership game and attackers adapted to each protected mechanism. Prespecified subgroup sensitivity, calibration and clinical operating thresholds guard against an apparent privacy gain purchased through poor rare-condition care (Table 4.1).

No reviewed study validates this complete clinical loop. Success would mean a credible paired reduction for the fixed baseline set, an upper bound on residual tail prevalence, no material worsening elsewhere, clinically acceptable subgroup utility, and a fair comparison to uniform protection under matched resources. Repeated seeds and independent reference populations are needed to separate a stable benefit from a favorable audit draw. Any formal claim must account for the data-dependent policy and patient adjacency [41,45,46].

9.5 Validation, Generalizability, and Re-Auditing of Patient-Level Protection

Validation applies the paired $\widehat{V}_p$ and $\widehat{B}_p$ definitions of Section 8 to a held-out audit: benefit in the fixed baseline set, residual exposure, and meaningful worsening outside it. Clinical shifts described in Section 8 and sample-level defense heterogeneity [15] motivate this test but do not constitute its completion.

A broad empirical claim requires adapted attacks with the same membership hypothesis, independent reference data, adequate low-FPR precision and repetitions across seeds and model families [2,14,48]. An attack left calibrated for the old mechanism can exaggerate benefit [15]. Disagreement among attacks can reveal different feasible adversaries rather than noise; retain their outcomes as a profile under Table 4.1.

External validation must separate patient membership from hospital, device and acquisition recognition. Subject-level ECG and hospital-dependency experiments show why site-matched non-members and reference populations matter [22,23]; shifts can strengthen or weaken an attack [25]. Independent cohorts, modalities, institutions and model updates should be tested before a general clinical claim, since multiple datasets inside one study are not independent replication of the proposed pipeline.

Privacy improvement is not clinically valid if it produces unacceptable predictive harm. Validation should apply the Section 8 criteria for overall and subgroup discrimination, calibration, and prespecified clinical operating points. Aggregate accuracy cannot substitute for subgroup validation, particularly for rare, underrepresented, and intersecting populations [7,31,47]. Prospective non-inferiority margins should be specified separately for overall utility and each clinically relevant group, with confidence intervals and subgroup sample sizes reported. The question is not only whether exposure falls, but whether the gain remains clinically acceptable for the populations whose data made the model useful.

Finally, validation must be treated as a continuing lifecycle rather than a permanent certification. The required cycle is

$$\text{audit} \rightarrow \text{protect} \rightarrow \text{independent re-audit},$$

followed by renewed auditing after changes in population, acquisition, preprocessing, model or protection mechanism. Re-auditing must measure absolute vulnerability among **all** patients: a new top-ranked patient alone is not evidence of worsened exposure [35]. Compare the targeted policy with a uniform baseline under matched utility, computation or genuinely comparable formal budgets. The aim is durable tail reduction without clinically important subgroup utility loss, not merely a reordered list.
